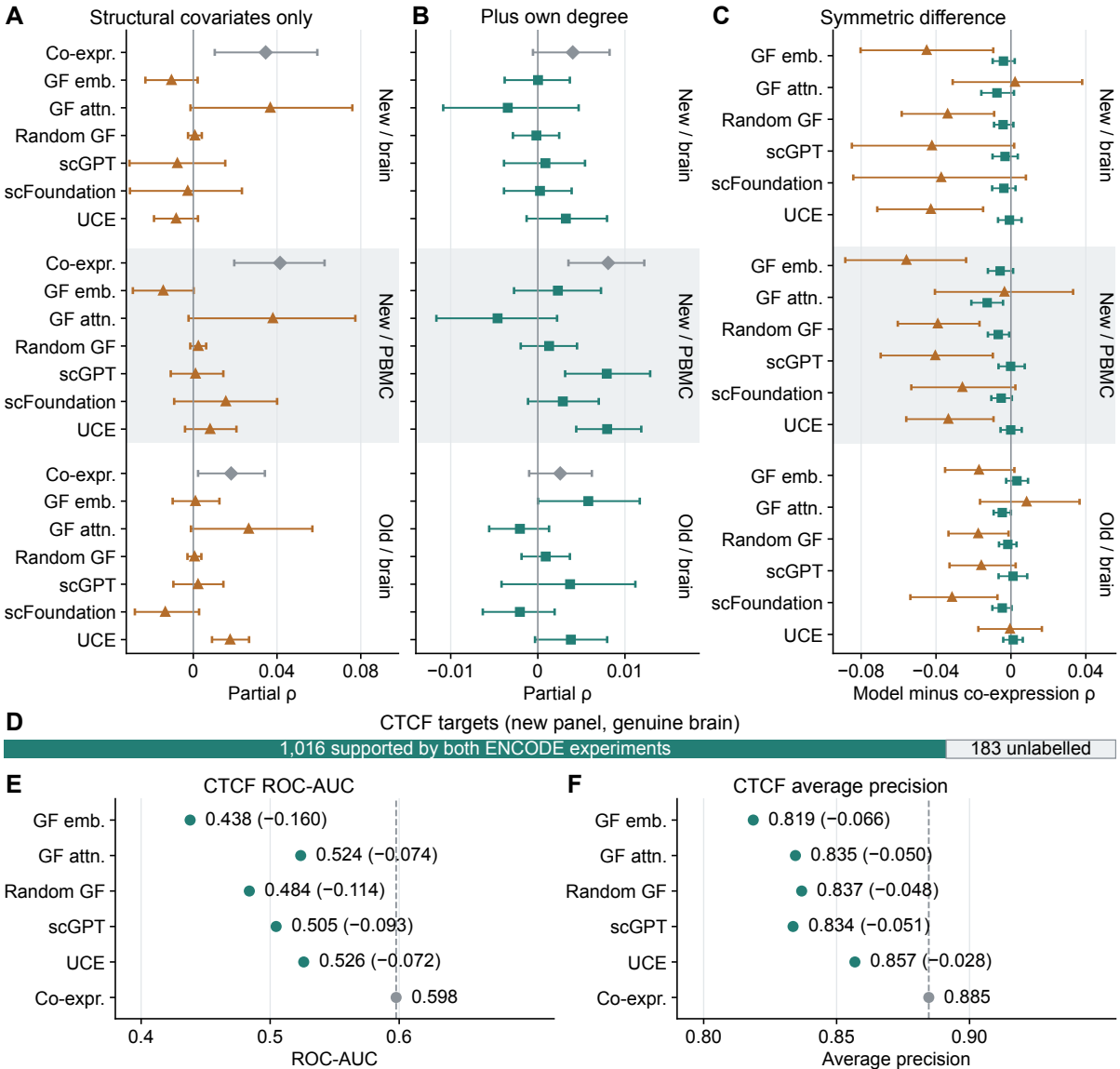
A
Structural covariates only
B
Plus own degree
C
Symmetric difference
Co-expr.
GF emb.
GF attn.
Random GF
scGPT
scFoundation
UCE
New / brain
New / PBMC
Old / brain
0
0.04
0.08
Partial ρ
−0.01
0
0.01
Partial ρ
−0.08
−0.04
0
0.04
Model minus co-expression ρ
D
CTCF targets (new panel, genuine brain)
1,016 supported by both ENCODE experiments
183 unlabelled
E
CTCF ROC-AUC
0.438 (−0.160)
0.524 (−0.074)
0.484 (−0.114)
0.505 (−0.093)
0.526 (−0.072)
0.598
0.4
0.5
0.6
ROC-AUC
F
CTCF average precision
0.819 (−0.066)
0.835 (−0.050)
0.837 (−0.048)
0.834 (−0.051)
0.857 (−0.028)
0.885
0.80
0.85
0.90
Average precision